

Porichoy (পরিচয়)

DPP-in-a-Box — every Bangladeshi garment gets a verifiable identity before the EU's 2027 Digital Product Passport wall.

Live pilot app: rudra496.github.io/porichoy

Team Porichoy · SUST IPE + BUET + UAP · Sylhet & Dhaka, Bangladesh

THE PROBLEM

From 2027, EU customs will auto-check Digital Product Passports on imported garments. No passport data → no EU order.

ESPR in force 18 Jul 2024 · textiles in the first Working Plan (Apr 2025) · textile delegated act expected ~2027.

Every factory already holds the proof — trapped in mixed Bangla/English Excels, ERP exports and paper trim cards. Global platforms are brand-side; nobody captures at the factory floor.

Most affected: small & mid-size exporters — and ≈ 4 million garment workers (BGMEA; majority women) whose market is Europe.

The dated wall: EU Digital Product Passport timeline (European Commission)



OUR SOLUTION

Our live pilot app turns a factory’s existing records into buyer-ready EU Digital Product Passports in four steps:

- 1 · INGEST — drop the factory Excel/CSV; parsed on-device. Raw data never leaves the factory.
- 2 · MAP — Bangla/English headers auto-mapped with confidence scores; a trained steward confirms.
- 3 · SCORE — DPP Readiness 0–100 per order (ESPR Art. 8 weights) + the exact gap list.
- 4 · PASSPORT — one click → QR; buyers scan and see the verifiable passport.

19/19 automated engine tests · SHA-256 provenance chain · Bangla + English · offline-tolerant

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DPP Readiness

Weighted share of EU-required fields you can already evidence

65

Compliance gaps

Shonali Textiles Ltd. (sample data) · 4 production orders

Bring records in

Upload a real factory Excel — or try the bundled messy sample first.

Upload records

Load sample data

Download readiness report (CSV)

Local-first: In this pilot your raw records never leave the factory device. Only the passport you choose to share is public.

Provenance chain live — 4 links · tip c4bdb74264b7d8c...

Production orders

PO	PRODUCT	DPP READINESS	MISSING	
PO-1001	Mens Polo Shirt S/S	90	gtin, hs_code	Open
PO-1002	Ladies Denim Jacket	90	gtin, hs_code	Open
PO-1003	Kids T-Shirt	45	gtin, hs_code, recycled_pct, facilities, certs, recycling	Open
PO-1004	—	35	gtin, hs_code, product_name, fiber_comp, recycled_pct, mfg_dates, care_info, recycling, substances	Open

Porichoy (পরিচয়) v2.0 — DPP schema v0.1 · local-first pilot · [Github](#)

16 Crossref-verified studies · facts sourced live · Built for Needle Innovation Challenge 3.0 · 2026

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Back

PO-1002 — Ladies Denim Jacket

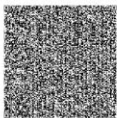
90

Buyer-ready

90/100 weight-points evidenced and validated.

Publish for the buyer

The passport snapshot rides inside the QR itself — opens on any phone, no cloud needed.



Open buyer view ↗

https://rudra496.github.io/porichoy/#/pp?id=eyJwb01kIjoiU0E8THTAuk1Is1nhJadVtVZ1cnlpb24101Iiw1JEl...

Copy link

Download passport (JSON)

Required attribute checklist

ATTRIBUTE	VALUE	SOURCE COLUMN	REGULATORY ANCHOR	WEIGHT
Production order / lot ID	PO-1002	পরিচয়	ESPR Art.8(a) unique identifier	10
Product code (GTIN/SKU)	missing	Add	ESPR Art.8(a) + GS1	5
HS tariff code	missing	Add	Customs linkage	5
Product description	Ladies Denim Jacket	Item Description	ESPR Art.8(b)	5

Live pilot app (demo data shown) · evidence page with 16 Crossref-verified studies · grant roadmap: OCR for paper trim cards, ERP/API connectors, multi-factory cloud sync



MARKET SIZE & OPPORTUNITY

A dated compliance wall creates a forced-adoption market:

≈ 4 million RMG workers (BGMEA; majority women) · world's #2 ready-made garment exporter · Europe is the biggest market · thousands of export factories must be DPP-ready inside the 2027–30 window.

Revenue math on our own pricing (BDT 2,000–8,000 / factory / month):

100 factories ≈ BDT 60 lakh / year · 1,000 factories ≈ BDT 6 crore / year — before brand-funded onboarding contracts, where buyers pay to bring their suppliers onto passports before the deadline.

≈ 4 million

RMG workers, majority women
(BGMEA)

#2 exporter

world RMG — Europe the biggest
market

2027–30

EU DPP phase-in; customs auto-checks

10 → 100+

factories: year-1 pilot → year-3 scale

Timing is the wedge: buyers already ask, budgets activate before the deadline — and no Bangladesh-built tool serves the factory floor today. Verified competitors are brand-side only.

BUSINESS & REVENUE MODEL

Three simple revenue lines:

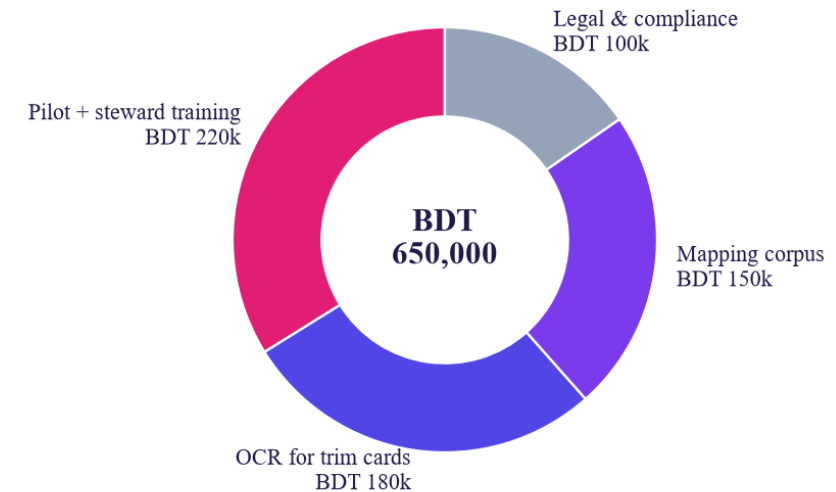
1. Factory subscriptions — BDT 2,000–8,000 / month by size (SME-priced, ~10× below enterprise platforms).
2. Onboarding & data-rescue — one-time fee: steward training + first ingest + mapping historic files.
3. Brand-funded supplier onboarding — buyers pay to make their supplier base passport-ready before 2027–30.

Unit economics: onboarding $\approx$ 2 person-days / factory · software gross margin $>$ 80% · replication is a template, not a project.

Pollination grant (BDT 650,000) funds the 10-factory pilot:

BDT 220k pilot + steward training · 180k OCR for paper trim cards · 150k mapping corpus (Bangla + ERP formats) · 100k legal & compliance.

Pollination grant allocation

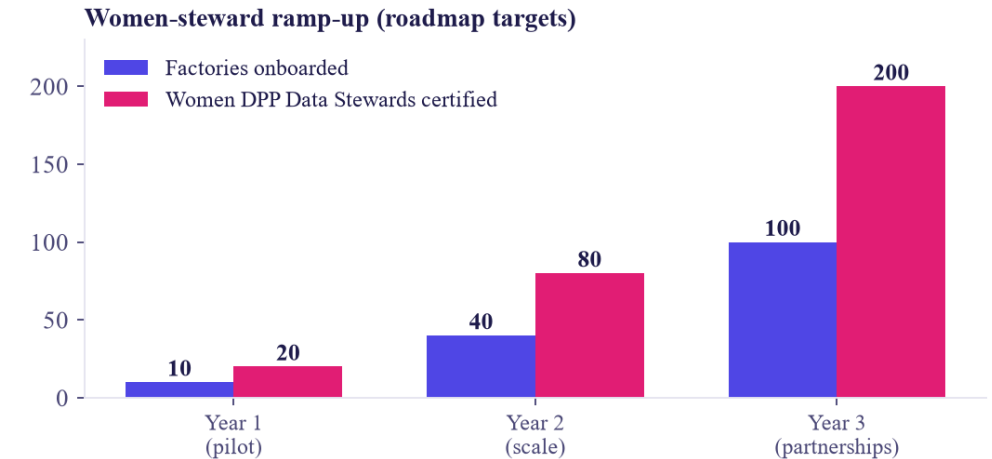


OUR IMPACT

Competitiveness: customs auto-checks make passport data order-critical — and compliance prep drops from days of file-hunting to under an hour per order.

Women: two women operators per factory trained & certified as DPP Data Stewards — formal, higher-skilled roles (20 in year 1 → 200 by year 3, targets). Order-eligibility protects a majority-women workforce ($\approx$ 4 million, BGMEA).

Circularity: recycled content, certificates and end-of-life routes become machine-checkable passport fields — circularity claims turn into verifiable data.



Women-steward ramp: roadmap targets. Photo: RMG factory floor — CC BY 2.0, Wikimedia Commons.

OUR TEAM



Rudra Sarker — Founder & full-time product engineer. Final-year B.Sc. Industrial & Production Engineering, SUST. Builds every part of Porichoy: the EU-aligned DPP schema, the Bangla/English mapping engine, readiness scoring and the provenance chain.

Verified awards: Q1 journal publication — Journal of Ethnopharmacology (Elsevier, 2026), 3rd author · Z.AI Global Ambassador · IEF Bangladesh Ambassador · Secretary, RoboSUST · GaoTek industrial internship.

Two IPE backgrounds ground us in factory reality; systems engineering makes the software ship. Women lead our steward program by design. No big-name advisers — a live product, public code and 19/19 passing tests speak instead.

Lutfay Homaira Islam

SUST · IPE

Factory Operations & Pilot Lead

Process mapping, factory data collection, readiness audits across the pilot factories

Priom Sarker

BUET · Naval Architecture & Marine Engineering

Systems & Technology Lead

Systems architecture, ERP/API integrations, technical review of the ingestion pipeline

Shuvo Kundu

University of Asia Pacific · EEE

Data & Research Lead

Dataset curation, Bangla mapping-corpus growth, literature & research support

Samia Hossain

Anwar Khan Modern University · CSE

Steward Training & Outreach Lead

Runs the women DPP Data Steward training program and factory outreach

Live: rudra496.github.io/porichoy · code: github.com/rudra496/porichoy · 19/19 engine tests · launched Aug 2026

Everything we claim is checkable — that is the product's whole point.

THANK YOU

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Live product: rudra496.github.io/porichoy · **Code:** github.com/rudra496/porichoy